

# Cloud Data Lifecycle

Lifecycle Flow:

- Create → Classify → Store → Process → Use  
→ Share → Archive → Retain → Delete

# Introduction to Cloud Data Lifecycle

- The cloud data lifecycle defines how data is managed from creation to destruction.

## Objectives:

- Ensure data security
- Maintain compliance
- Optimize storage cost
- Enable scalability
- Improve governance
- Reduce risk

# 1. Data Creation / Ingestion

- Definition: Data is generated or collected.

## Sources:

- Applications
- APIs
- IoT devices
- Logs
- Sensors
- User uploads

## Key Focus:

- Data quality
- Validation
- Schema design
- Integrity

## 2. Data Classification

- Definition:
- Categorizing data based on sensitivity.

### Types:

- Public
- Internal
- Confidential
- Restricted

### Why Important:

- Security policies
- Access control
- Compliance mapping
- Risk management

# 3. Data Storage

## Storage Models:

- Object storage
- Block storage
- File storage
- Databases

## Design Principles:

- Durability
- Availability
- Redundancy
- Replication
- Geo-distribution

# 4. Data Processing

Purpose:

- Transform raw data into useful information.

Processes:

- ETL/ELT
- Batch processing
- Stream processing
- Analytics
- AI/ML training

Outcomes:

- Clean data
- Structured data
- Insights

# 5. Data Usage / Access

## Consumers:

- Applications
- Dashboards
- Reports
- APIs
- AI systems

## Controls:

- IAM
- RBAC
- Encryption
- Monitoring
- Audit logs

# 6. Data Sharing / Distribution

## Methods:

- APIs
- Cross-account access
- Data lakes
- Integrations

## Risks:

- Data leakage
- Over-permissioning
- Shadow access

## Controls:

- Trust boundaries
- Governance rules



# 7. Data Archival

## Definition:

- Moving inactive data to low-cost storage.

## Use Cases:

- Historical records
- Legal archives
- Compliance data

## Benefits:

- Cost reduction
- Long-term retention

# 8. Data Retention

## Definition:

- Keeping data for required duration.

## Drivers:

- Legal policies
- Regulatory rules
- Business needs

## Examples:

- Logs – 90 days
- Financial data – 7 years

# 9. Data Deletion / Destruction

## Definition:

- Permanent removal of data.

## Methods:

- Secure deletion
- Cryptographic erasure
- Automated lifecycle rules

## Purpose:

- Compliance
- Privacy
- Risk reduction

# Cloud Data Lifecycle Architecture View

Layers:

- Ingestion Layer
- Storage Layer
- Processing Layer
- Access Layer
- Governance Layer
- Security Layer

# Conclusion

Cloud Data Lifecycle ensures:

- Controlled data flow
  - Secure data handling
  - Compliant data management
  - Efficient cloud operations
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- It is a foundation for cloud governance and enterprise data architecture.